

Deep learning approach for automated ‘Kent’ mango maturity grading in compliance with Peruvian standards

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ABSTRACT

Deep learning, particularly convolutional neural networks (CNNs), has significantly advanced automated fruit classification based on image analysis. However, accurate classification of *Mangifera indica* L. remains challenging due to high variability in external appearance and the subjectivity of visual maturity assessment. Misclassification contributes to post-harvest losses, reduced market value, and inconsistencies in quality control. This study develops a CNN-based model for classifying ‘Kent’ mangoes according to the Peruvian Technical Standard (NTP) 011.025:2023. A dataset of 603 labelled images was used to optimise the CNN architecture, systematically evaluating convolutional and pooling layers, image resolution, and training cycles. The optimised model, trained on 32×32 pixel images, achieved 96.04 % classification accuracy, 90.91 % recall, and an F1-score of 93.57 %. To validate model robustness, 5-fold cross-validation demonstrated minimal accuracy variation (± 0.5 %), while external evaluation achieved 95.8 % accuracy, confirming its real-world applicability. The lightweight single-layer CNN ensures scalable, low-cost implementation for automated sorting systems, reducing computational demands while enhancing classification efficiency. These findings establish deep learning as a viable and cost-effective solution for post-harvest fruit classification, ensuring greater consistency in quality control and supporting sustainable agricultural practices.

1. Introduction

Mangifera indica L. is one of the most commercially traded tropical fruits, playing a significant role in global agricultural economies [1]. It is cultivated in over 90 countries and ranks fifth in worldwide fruit production, following bananas, apples, grapes, and citrus

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fruits [2–3]. The global demand for mangoes has increased steadily over the last decade due to their nutritional richness, which includes a high content of fibre, vitamins (A, C, and E), and bioactive compounds with antioxidant and anti-inflammatory properties [4–5]. These characteristics, combined with the fruit’s unique flavour, texture, and versatility, have contributed to a growing international market, particularly in Latin America, where Mexico, Brazil, and Peru are the primary exporters [6].

Despite its commercial significance, mango post-harvest management presents considerable challenges, particularly concerning fruit perishability, variability in maturity stages, and susceptibility to mechanical damage. The fruit’s shelf-life is significantly affected by physical factors such as compression, bruising, and sunburn, leading to deterioration in colour, texture, and nutritional quality [7]. Mangoes are climacteric fruits, meaning they undergo rapid biochemical transformations post-harvest, including increased acidity, enzymatic degradation, mesocarp softening, and cell wall breakdown [8]. These physiological changes alter both internal and external characteristics, impacting not only marketability but also storage feasibility, transport logistics, and consumer satisfaction [9]. Furthermore, inconsistencies in maturity assessment can impact export logistics, as improperly classified fruits may fail to meet international market standards, leading to economic losses and increased food waste.

A major determinant of post-harvest quality control is mango maturity assessment, as it affects market value, storage potential, and overall fruit appeal [10]. The most commonly used maturity indicator is external peel colour, yet this method is inherently subjective and unreliable. External hue does not always correlate with internal maturity, as different cultivars exhibit varied pigmentation patterns [11]. At harvest, mango flesh may appear pale yellow, while the peel can range from yellow-green to reddish-green. Upon full maturation, the mesocarp transitions to yellow-orange, whereas the peel develops yellow-reddish pigmentation, introducing complexity in classification (see Fig. 1) [12–13]. This high degree of visual variability complicates manual sorting operations, making classification errors frequent and increasing post-harvest waste [14].

Given the high cost of extending mango shelf-life through cold chain logistics, there is a pressing need for more precise and scalable classification solutions. However, current methods rely on manual assessment, which is prone to human error, fatigue, and inconsistencies in evaluation criteria. Given that the export market requires strict quality control standards, these inconsistencies result in economic losses and compromised fruit quality for international markets [16]. Therefore, there is a pressing need for automated, standardised classification methods capable of ensuring accuracy, efficiency, and consistency.

In response to these challenges, machine vision systems, integrated with artificial intelligence, have emerged as a promising alternative for fruit sorting applications [17]. These systems leverage image processing techniques to analyse colour, texture, and shape, providing non-invasive, objective evaluations that improve classification outcomes [18]. Machine vision approaches for mango maturity classification have included pattern recognition via circular Hough transforms [19], adaptive thresholding for colour segmentation [20], surface texture analysis using feature extraction techniques [21], and histogram-based segmentation for enhanced colour assessment [22].

Traditional machine learning models, such as support vector machines (SVMs) and structure-based matching, have demonstrated moderate classification accuracy in fruit sorting applications [17]. However, recent advances in deep learning, particularly CNNs, have significantly outperformed classical approaches, achieving higher accuracy and greater adaptability across different fruit cultivars and conditions [23].

CNNs function by hierarchically extracting image features, progressively refining spatial and spectral representations of an image for optimal classification [24]. The architecture consists of:



Fig. 1. Mango harvesting process in Peru: manual selection and handling.

1. Convolutional layers: Detect fundamental edges, textures, and colour gradients.
2. Pooling layers: Reduce computational complexity while retaining essential visual features.
3. Fully connected layers: Synthesise extracted information into classification outputs [23].

Unlike conventional models, CNNs eliminate the need for manual feature engineering, allowing for direct learning of relevant visual patterns from labelled datasets [25]. Furthermore, transfer learning techniques enable CNNs to adapt pre-trained architectures to specific classification tasks, reducing the need for extensive datasets while maintaining high accuracy [26]. Compared to SVM and K-nearest neighbour (KNN) models, CNNs offer greater robustness to variations in lighting, fruit shape, and texture, making them particularly effective for classifying climacteric fruits with high inter-class variability [27]. Additionally, CNN-based models can generalise across different phenological stages, making them ideal for industrial-scale sorting applications [28].

This research aims to develop a highly efficient CNN-based computational model for automated mango classification, aligned with the NTP 011.025:2023 “Fresh Mango”. A dataset comprising 603 labelled images was used to train and optimise the CNN architecture, systematically evaluating different configurations to balance classification accuracy and computational efficiency. The proposed model was implemented in Python with TensorFlow, offering a high-accuracy, low-latency solution for post-harvest fruit sorting at an industrial scale.

2. Materials and methods

2.1. Mango sample collection and dataset construction

This study utilises a dataset of 603 labelled images of ‘Kent’ mangoes, sourced from three key producing regions in northern Peru: Piura, La Libertad, and Lambayeque. The dataset was curated in collaboration with the National Association of Organic Producers of Peru (ANPE PERÚ) to ensure a representative sample of industry conditions. Following the NTP 011.025:2023 “Fresh Mango” [29], mangoes were categorised based on their physiological development:

- Green: Immature mangoes with firm flesh and a predominantly green external colour.
- Mature: Mangoes at the optimal stage for consumption, characterised by yellow flesh and skin exhibiting yellow to reddish variations.

The dataset consists of 265 images of mature mangoes and 338 images of green mangoes, maintaining the original class distribution ratio. To facilitate model evaluation, the dataset was randomly divided into 301 images for training and 302 images for testing, ensuring a balanced distribution that accurately reflects real-world production conditions.

2.2. Acquisition architecture and preprocessing pipeline

Image acquisition was conducted in a semi-controlled environment to minimise colour distortions and standardise dataset consistency under varying ambient conditions. A controlled white balance calibration was performed before each capture session, and ambient illumination was standardised at 5500 K using a neutral daylight LED source to enhance image fidelity. Shadows and reflections were mitigated using a diffused lighting system, ensuring uniformity across samples. Natural light close to the neutral spectrum (5500 K) was selected, as it minimally alters the tonal quality of images, reducing the introduction of colour distortions [30]. To prevent complications in image processing, direct exposure to sunlight was avoided [31]. The images were captured using a LifeCam HD-3000 camera (Redmond, WA, USA), equipped with a CMOS sensor chosen for its high sensitivity and low noise levels [32]. The camera was stabilised on a tripod to ensure high capture quality. Each image was stored in JPEG format with a spatial resolution of 1200 × 800 pixels, a tonal depth of 8 bits per channel (24 bits per pixel), and a spatial resolution of 0.02 mm/pixel, covering the three RGB channels. Image evaluation was conducted on a Lenovo Legion Intel® Core™ i7–10750H laptop (Lenovo, China). This setup, simulating real-world harvesting environments, ensures that the model remains highly reproducible in practical agricultural applications where environmental control is limited.

2.3. Analysis and validation of datasets

Upon arrival, all mangoes were photographed under controlled conditions and classified based on their physiological maturity, adhering to the NTP 011.025:2023 “Fresh Mango”. The classification criteria remained consistent with Section 2.1, ensuring standardisation throughout the dataset. The dataset was subsequently structured into training and test subsets, maintaining a balanced

Table 1
Dataset distribution for mango maturity classification: training and testing splits.

Type	Total	Training	Test
Mature	265	132	133
Green	338	169	169
Total	603	301	302

representation of both maturity categories (Table 1). This methodological approach ensures the robustness of model evaluation, minimising potential biases arising from dataset imbalances.

A resolution of 32×32 pixels was selected to optimise computational performance while maintaining high classification accuracy. Although higher resolutions (e.g., 64×64 or 128×128) can capture more detail, they increase computational complexity and the risk of overfitting. Prior research supports the effectiveness of lower resolutions for similar classification tasks, as they sufficiently capture essential fruit features like colour and shape [33]. Experimental trials confirmed that increasing resolution beyond 32×32 pixels did not yield substantial accuracy improvements but significantly increased training time.

To enhance model generalisation, data normalisation was applied to scale pixel values between 0 and 1. Additionally, Data Augmentation techniques were implemented, including random rotations, brightness adjustments, zooming, and flipping, to increase dataset variability and improve model robustness in different lighting conditions [34]. These transformations expanded the dataset's representativeness, allowing the model to learn distinctive mango characteristics under varied environmental settings [35]. To evaluate the uniformity of image acquisition and validate colour distribution, histograms were generated for each of the three RGB channels (Fig. 2).

This histogram-based analysis lends validation to the efficacy of RGB-based classification, demonstrating a decline in green values concomitant with an intensification of red values as mangoes undergo the process of maturation. These trends are consistent with those observed in previous studies on colour transitions in climacteric fruits, thus confirming the reliability of hue-based feature extraction as a reliable indicator of mango maturity [36].

2.4. CNN architecture and optimization

The computational model was designed using a standard CNN architecture, widely applied in image classification tasks due to its ability to autonomously extract spatial features [36]. Implemented in Python 3.11 (Wilmington, USA) [37] with the Keras and TensorFlow libraries, the model utilised Google Colab for cloud-based computational resources, following best practices in deep learning applications [38].

The CNN processes $224 \times 224 \times 3$ pixel input images, a resolution selected through preliminary trials to balance classification accuracy and computational efficiency. While lower resolutions (e.g., 32×32 pixels) minimized processing demand, higher resolutions improved feature extraction at increased computational cost. The model applies a 3×3 kernel with 32 filters in the convolutional layer, using "same" padding to preserve spatial dimensions. A stride of 3 pixels optimizes spatial detail retention while maintaining efficiency. The ReLU activation function introduces non-linearity, preventing vanishing gradients and enhancing feature extraction. Extracted feature maps undergo max-pooling with a 2×2 kernel and a stride of 2 pixels, reducing dimensionality while preserving essential spatial patterns, mathematically defined as $f(I_{11}, I_{12}, I_{21}, I_{22}) = \max(x, y)$, selecting the highest pixel intensity within a defined region.

To enhance training stability, batch normalization is applied after each convolutional layer, mitigating internal covariate shifts and

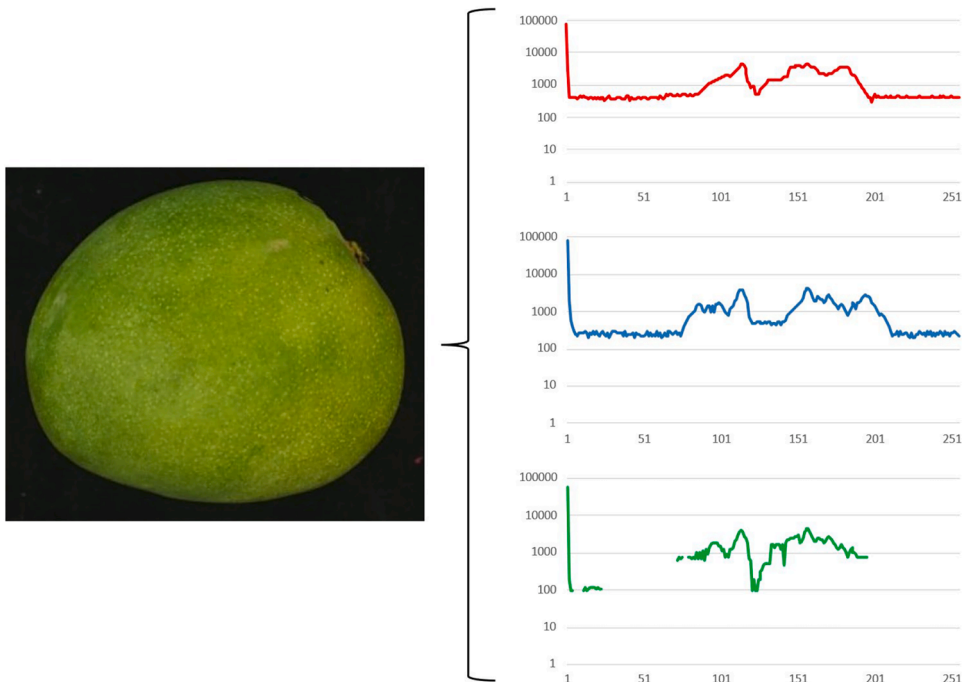


Fig. 2. RGB histogram analysis of mango maturity stages.

accelerating convergence. The pooled feature maps are flattened into a unidimensional vector and passed through fully connected layers to integrate learned features. A 50 % dropout rate prevents overfitting, ensuring the network generalizes rather than memorizes training data [39]. The output layer, consisting of two neurons (mature and green classes), employs the Softmax function to generate probability distributions. Bayesian optimization was used to fine-tune filter sizes, activation functions, and dropout rates, resulting in a final configuration with 32 filters, a 3×3 kernel, ReLU activation, and 50 % dropout, balancing feature extraction and regularization. Training was conducted over 100 epochs with a batch size of 32, using categorical cross-entropy loss, a standard metric for multi-class classification [40]. The Adam optimizer was selected for its adaptive learning rate adjustments, enabling fast and stable convergence [41]. Although early stopping was considered, it was excluded to allow for full convergence analysis.

The CNN architecture follows a hierarchical feature extraction approach. As illustrated in Fig. 3, convolutional layers capture spatial patterns such as edges, textures, and colour gradients. The pooling layers then perform dimensionality reduction, preserving essential features while reducing computational overhead. Finally, the fully connected layers integrate extracted features to classify mangoes by maturity stage. The Softmax function ensures accurate class differentiation between “mature” and “green” mangoes. This optimized CNN architecture achieves a balance between classification accuracy and computational efficiency, making it suitable for real-time automated post-harvest fruit sorting.

2.5. Model evaluation

The model’s performance was assessed using accuracy, recall, and F1-score, standard metrics in image classification tasks [42]. Accuracy represents the proportion of correct predictions over the total number of predictions. However, in datasets with imbalanced classes, accuracy alone may be misleading, making recall and F1-score more reliable indicators of classification performance. Recall evaluates the model’s ability to correctly identify mature mangoes, defined by Eq. (1):

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (1)$$

Where TP (True Positives) represents mangoes correctly classified as mature, while FN (False Negatives) denotes mature mangoes misclassified as green. A high recall score indicates a low false negative rate, which is crucial in post-harvest classification to prevent premature rejection of marketable mangoes.

The F1-score, a harmonic mean of precision and recall, is particularly valuable in binary classification tasks with imbalanced classes, as it provides a single metric that balances the impact of both false positives and false negative are accounted for in model evaluation. It is calculated as follows (Eq. (2)):

$$\text{F1 - Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2)$$

This metric is widely used in deep learning studies as it provides a balanced assessment of classification performance, particularly when precision and recall values differ significantly [43].

To further analyse model effectiveness, a confusion matrix was constructed, detailing True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). This matrix provides a comprehensive performance breakdown, allowing insights into

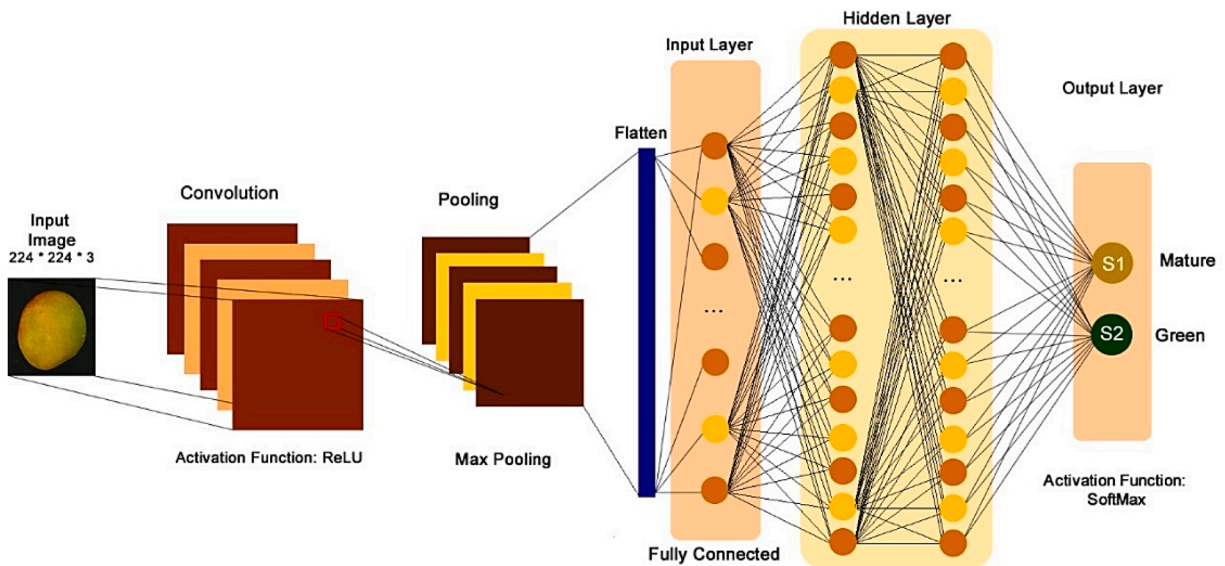


Fig. 3. CNN architecture for mango classification.

misclassification trends and guiding potential refinements in model tuning and dataset augmentation strategies.

Additionally, a 5-fold cross-validation strategy was implemented to assess model stability and robustness across multiple dataset partitions. This method ensures that each sample is used for both training and testing, providing a more reliable estimate of generalisation performance while mitigating potential biases caused by dataset variability [44].

3. Results and discussions

This study presents the development of a computational model for mango classification that is based on a CNN architecture. The model was trained on a dataset of 603 images, and various configurations were evaluated to optimise its performance through parameterisation. Adjustments were made to five models, and various combinations of layers (convolutional and reduction) were tested. Fifteen training epochs were run with a mini-batch size of 15 images. Model accuracy was calculated on both the test and training datasets.

3.1. Effect of convolutional and pooling layers

In accordance with Kumar et al., [45], a convolution and pooling layer are recommended. The number of levels has varied, as shown in Table 2.

Fig. 4 shows that 4 types of layer configurations were evaluated, each including a convolution block and a reduction block. The best performance was achieved with type 1, which consists of 1 convolution and reduction block, achieving an accuracy of 94.94 %.

Our study, which employed a CNN with only 1 convolutional layer and 1 pooling layer, reached a classification accuracy of 96.04 %. Compared to recent models using ResNet and InceptionV3 architectures for fruit classification, our lightweight CNN design achieves a similar classification accuracy with significantly lower computational cost., [46] employed an 8-layer CNN for fruit categorization, attaining 95.67 % accuracy, while our streamlined model reaches 96.04 %, highlighting the efficiency of minimalistic network architectures in constrained environments. This result is notably close to the 95.67 % obtained by Wang and Chen [46], who used a considerably more complex 8-layer CNN architecture for fruit classification. This comparison highlights that, despite its simplicity, our CNN architecture demonstrates strong performance, optimising computational resources and training time without significantly compromising accuracy. Additionally, the work of Ma et al., [47] demonstrated the performance of CNNs in mango classification, achieving a mean square error of 0.016 and an R^2 of 0.99, further supporting the viability of simpler models such as ours in delivering high-precision results. These findings suggest that, while our CNN structure is more straightforward, its competitive performance underscores its potential for future optimisations that could further increase accuracy without increasing the model's complexity. In comparison with prior studies utilising complex architectures or advanced techniques such as ensemble learning, our simplified model achieves competitive results with reduced computational costs, making it especially suitable for industrial contexts where simplicity and performance are critical.

3.2. Spatial resolution of the input images

The images were initially captured at a resolution of 1280×800 pixels in three RGB channels. The images were subsequently resized to different resolutions to assess their impact on classification accuracy, as illustrated in Fig. 5.

The classification accuracy was evaluated across four different input image resolutions: 32×32 pixels, 64×64 pixels, 150×150 pixels, and 1280×800 pixels. The results demonstrate that the 32×32 pixel resolution achieves the highest accuracy, with a score of 96.12 %, outperforming the other resolutions tested. Notably, there was a slight decrease in accuracy of 1.03 % and 1.18 % at the 64×64 and 150×150 pixel resolutions, respectively, highlighting the effectiveness of lower resolutions for CNNs in this context. This finding aligns with previous research, which has shown that smaller input image dimensions are more effective for CNN-based classification, particularly when the original images are modest in size [48–50]. The high accuracy at 32×32 pixels is consistent with the findings of Kumar et al., [45], who reported that a CNN achieved near-perfect classification at even lower resolutions. The selected resolution is sufficient to capture the key features necessary for accurate classification while avoiding the computational burden of higher resolutions. This underscores that reducing resolution can enhance both classification accuracy and computational

Table 2
Influence of Convolutional and Pooling Layers on CNN Accuracy.

Type	Number of Layers	Accuracy
1	1 Convolution – 1 Pooling	94.94 %
2	1 Convolution – 1 Pooling	89.95 %
3	1 Convolution – 1 Pooling	93.04 %
4	1 Convolution – 1 Pooling	90.90 %

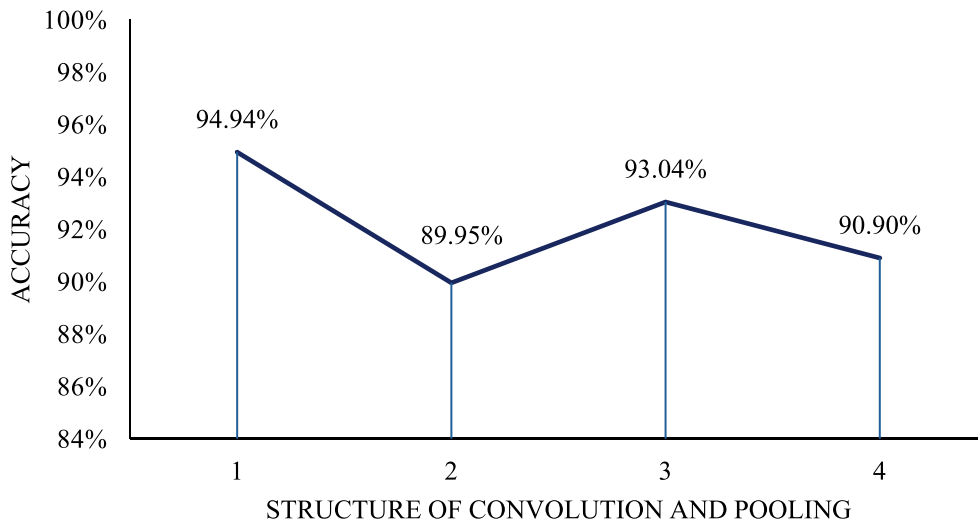


Fig. 4. Effect of convolutional and pooling layers on classification accuracy.

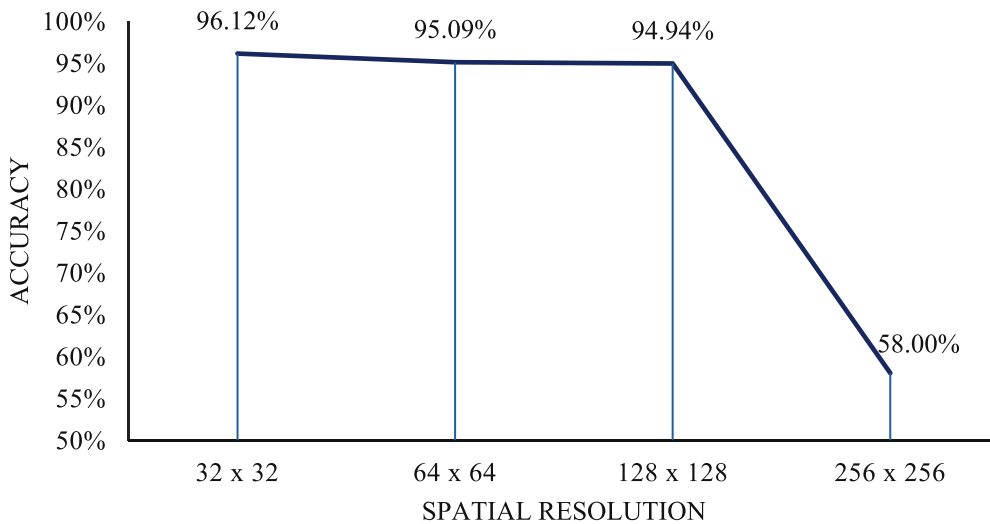


Fig. 5. Impact of input image resolution on CNN classification performance.

efficiency, especially in fruit classification tasks.

3.3. Number of epochs used in training and validation

The training of the CNN is conducted over multiple repetitive cycles known as epochs, during which the model adjusts its parameters to enhance classification accuracy and mitigate issues such as overfitting, as illustrated in Fig. 6.

The training of the CNN is conducted over multiple repetitive cycles known as epochs, during which the model adjusts its parameters to enhance classification accuracy and mitigate issues such as overfitting, as shown in Fig. 6. The correlation between the number of epochs and the accuracy achieved during the training and validation phases of the CNN model is depicted in this figure. Initially, after 10 cycles, an accuracy of 96.12 % was observed. With an increase in the number of epochs to 20 and 30, a slight improvement was noted, with accuracies reaching 96.99 % and 96.91 %, respectively. However, the model demonstrated 100 % accuracy at epochs 60 and 160, indicating that these points may represent the optimal performance range. It is crucial to consider the potential risk of overfitting, where the model becomes overly tailored to the training data, thereby impairing its ability to generalise to new data. Overfitting is evidenced when the analysis results plateau at 100 % accuracy beyond the 60th epoch, suggesting that the model may have learned specific features of the training data rather than generalising useful patterns. This finding aligns with previous

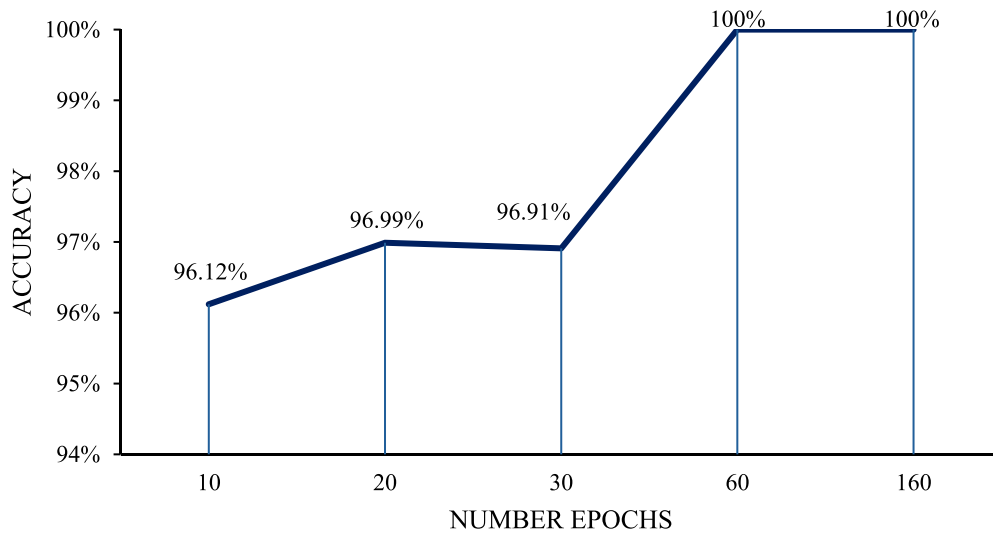


Fig. 6. Model convergence analysis across different epochs.

studies, such as that of Kumar et al., [45], who reported a high error rate during the initial epochs, followed by a significant reduction between epochs 9 and 40, where the error rate is minimal without signs of overfitting. Consequently, to minimise the risk of overfitting and optimise computational performance with a minimum error rate (between 3.88 % and 0 %), it is recommended that the model be trained within a range of 10–60 epochs while carefully monitoring its performance on a validation set to ensure that its generalizability is not compromised.

3.4. Parametrisation of the CNN model

The performance of the CNN model was evaluated through training and validation metrics, with Figs. 7 and 8 illustrating the evolution of accuracy and loss throughout the training process. The final configuration for mango image classification was determined by systematically evaluating different architectural settings. The optimised model consisted of a single convolutional layer, a pooling

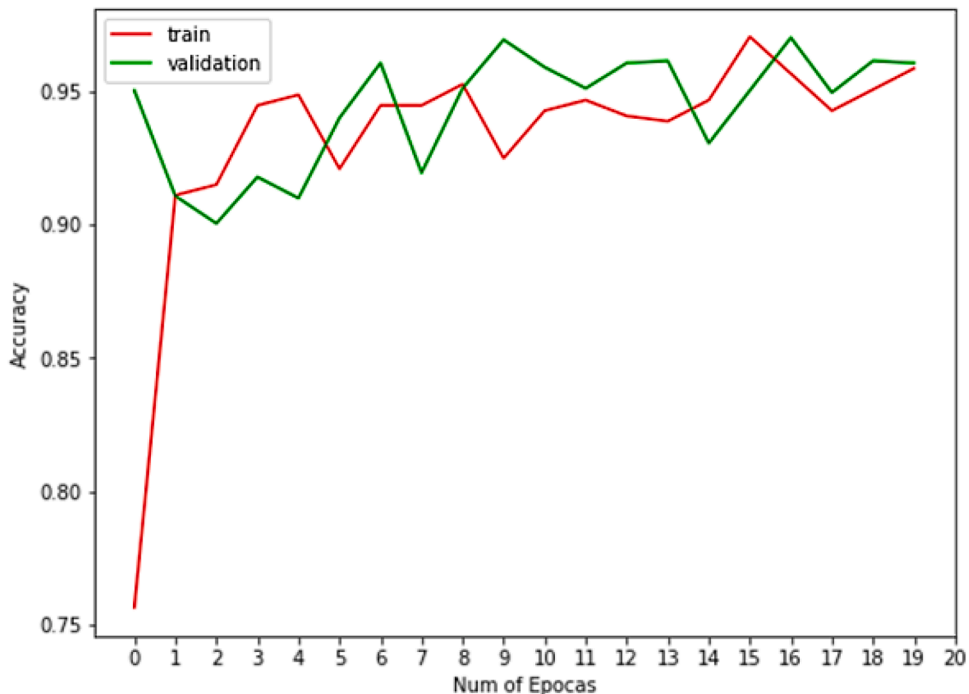


Fig. 7. CNN training and validation accuracy trends.

layer, an input resolution of 32×32 pixels, and a training duration of 20 epochs, striking a balance between computational efficiency and classification performance.

Fig. 7 presents the training and validation accuracy curves. The rapid increase in training accuracy during the early epochs indicates efficient feature extraction and pattern recognition within the dataset. Although the validation accuracy exhibits slight fluctuations, it follows a consistent upward trend, eventually converging with the training accuracy. This suggests that the model effectively generalises to unseen data without significant overfitting.

Fig. 8 illustrates the evolution of training and validation loss, where the training loss sharply decreases within the first few epochs, a characteristic of well-optimised learning processes. The validation loss, while exhibiting minor variability, follows a similar declining trend, stabilising at lower values in later epochs. The close alignment of both loss curves indicates that the model maintains a strong balance between learning from training data and adapting to new data, confirming its robustness in classification tasks.

The developed model achieved an accuracy of 96.04 %, demonstrating high effectiveness in mango classification based on colour tone and maturity. This success can be attributed to the efficiency of the backpropagation algorithm, which optimises digital image processing through convolutional and pooling layers. The model's streamlined architecture, requiring only 20 training epochs, ensures rapid deployment in industrial settings while maintaining computational efficiency. This design is particularly advantageous for resource-limited environments where high-performance deep learning solutions are often impractical.

A comparative analysis with recent studies highlights the model's effectiveness. Zhang et al. [44], Tang et al. [28], and Xiao et al. [51] reported classification accuracies between 92 % and 99.8 %, while Sakib et al. [52] achieved 100 % accuracy when classifying 25 fruit categories. However, many of these models rely on highly curated datasets and single-variable input features, limiting their practical applicability in real-world industrial conditions. In contrast, the dataset used in this study was acquired under natural conditions without sample manipulation, ensuring a more realistic representation of industrial mango processing environments. This approach aligns with previous efforts by Bhargava and Bansal [53] and Casado-García et al. [54], who incorporated natural colour, shape, and texture variations to enhance model adaptability, as similarly explored by Sa et al. [55] and Javaid et al. [56].

Despite recent advancements in automated fruit classification, certain deep learning models face limitations in real-world deployments due to their computational intensity and reliance on extensive hardware infrastructure. Rizwan and Hakim [57] implemented the Inception v3 model, achieving 99.2 % accuracy, but the high processing requirements reduce its practicality for industrial environments. Similarly, Gururaj et al. [58] combined CNNs with Random Forest, obtaining 93.23 % accuracy in variety identification and 95.11 % in quality classification, but their method involves substantial computational overhead. In contrast, the CNN model developed in this study achieves competitive classification accuracy with significantly reduced computational demands, making it particularly well-suited for industrial-scale mango sorting applications [59].

Although advanced architectures such as those proposed by Zoph et al. [60] enable highly scalable deep learning solutions, this study prioritises efficiency and practicality, ensuring that the model remains viable under resource-constrained conditions without sacrificing classification accuracy. Nithya et al. [61] achieved 98 % accuracy in defect detection using CNNs, but their approach

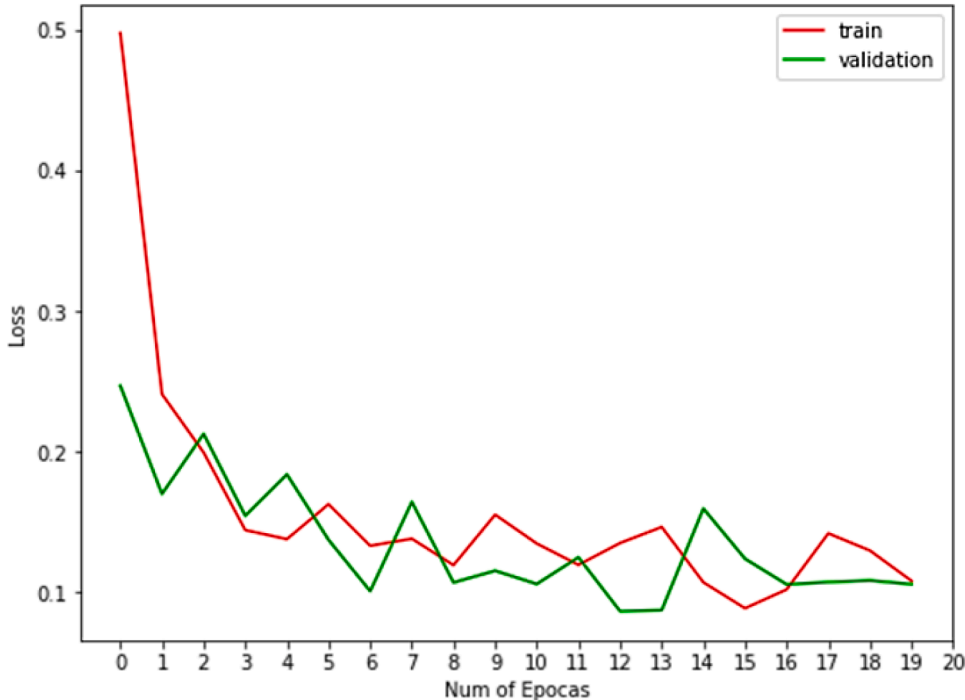


Fig. 8. Training vs validation loss evolution during CNN learning.

focuses solely on surface defects, disregarding maturity assessment, which is a critical factor in mango shelf-life prediction. Additionally, Tai et al. [62] explored ensemble learning techniques (bagging, boosting, and stacking), reaching classification accuracies exceeding 98 %, but these methods generally require higher computational resources, potentially restricting large-scale implementation.

This study demonstrates that a simplified CNN framework can achieve high classification accuracy (96.04 %) while maintaining computational efficiency, making it a viable solution for automated mango sorting systems in post-harvest processing facilities.

3.5. Model performance

To comprehensively evaluate the model's performance, a confusion matrix was constructed using the predictions from the test dataset (Fig. 9). This matrix provides a structured representation of classification outcomes, detailing TP, TN, FP, and FN, which are fundamental for assessing model effectiveness, recall, and F1-score in binary classification tasks [63,64]. The high recall (90.91 %) ensures accurate identification of mature mangoes, minimising misclassification risks in automated sorting. The strong F1-score (93.57 %) further reflects the model's balanced precision and recall, reducing classification errors that could lead to economic losses in post-harvest operations [43].

Despite its high accuracy, misclassifications primarily occur during transitional maturation stages, where spectral similarities between mature and immature mangoes increase. These errors have been documented in previous fruit classification studies, where subtle hue and textural variations complicate differentiation [65]. Hybrid CNN-Transformer architectures have demonstrated superior capability in extracting both local and global spatial patterns, improving classification performance in scenarios with high intra-class variability [66,67]. Implementing such architectures in mango maturity classification could further optimise feature representation while maintaining computational efficiency.

A 5-fold cross-validation was conducted, revealing minimal accuracy variation (± 0.5 %), which confirms the model's stability across different dataset partitions and its generalisability to unseen data. The model was also tested on an external dataset, achieving 95.8% accuracy, further validating its adaptability to real-world deployment in industrial sorting systems [68].

Beyond traditional classification approaches that focus solely on surface defects, this model simultaneously evaluates maturity stage and defects [56,69], improving its applicability in automated sorting systems. The integration of deep learning and computer vision enables efficient CNN parameterisation, ensuring stable classification performance while minimising overfitting through dropout regularisation and data augmentation [70,71]. These findings align with low-resolution image-based classification studies, demonstrating that essential maturity differentiation features can be extracted with minimal computational cost [15,72].

Although the 32×32 pixel resolution has proven effective, it may limit the detection of finer textural and chromatic details, affecting classification precision in subtle maturity transitions. Super-resolution techniques, such as the Superpixel Token Interaction method, could enhance CNN feature extraction, but must be balanced against computational efficiency for scalability [73]. Future research should evaluate 64×64 pixel resolutions and incorporate geographically diverse datasets to improve generalisation across different environments and mango varieties.

Cross-validation strategies, particularly 5-fold cross-validation, are widely recognised for reducing overfitting and enhancing model reliability [74]. Expanding these validation methods and testing the model in varied production environments and lighting conditions could further assess its robustness. Additionally, implementing higher-resolution imaging and adaptive learning techniques may enhance classification performance and scalability.

While increasing dataset diversity and resolution may improve accuracy, these modifications must be weighed against computational constraints [75]. Nonetheless, the proposed CNN model remains a robust, scalable, and efficient solution for automated mango classification, providing a foundation for future advancements in machine vision-based fruit sorting systems.

4. Conclusion

This study presents an optimised CNN-based model for mango maturity classification, achieving 96.04 % accuracy, 90.91 % recall, and an F1-score of 93.57 %, confirming its reliability in distinguishing maturity stages. Its lightweight, single-layer architecture enables real-time classification with reduced computational cost, making it suitable for large-scale industrial sorting systems where computational efficiency is critical.

The model's robustness was validated through 5-fold cross-validation, with minimal accuracy variation (± 0.5 %), and external testing achieved 95.8 % accuracy, demonstrating its adaptability to real-world post-harvest operations. These results highlight the potential of deep learning to improve quality control, reduce classification errors, and optimise supply chain efficiency by minimising food waste and post-harvest losses.

Although dropout regularisation mitigated overfitting, additional refinements through early stopping, adaptive learning, and transfer learning could further enhance model generalisation across diverse production conditions. Future research should investigate the integration of high-resolution imaging, hyperspectral techniques, and hybrid CNN-transformer architectures to improve classification accuracy and adaptability across different mango cultivars, environmental conditions, and ripening stages.

Despite these challenges, the proposed model provides a scalable, cost-effective, and high-performance solution for automated fruit classification, supporting AI-driven agro-industrial automation and contributing to the development of more sustainable and intelligent post-harvest processing systems.

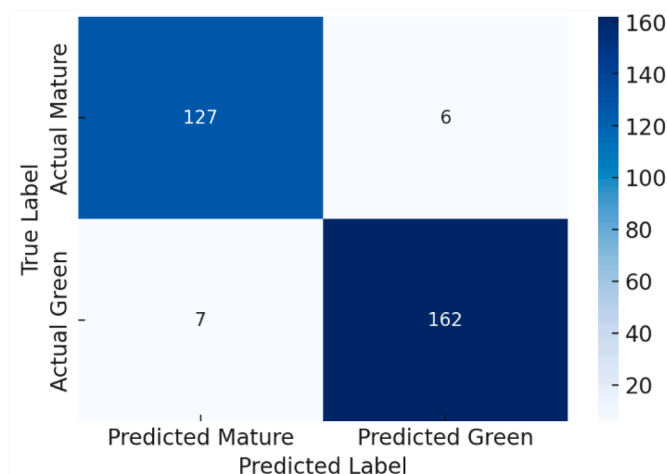


Fig. 9. Confusion matrix analysis of mango maturity classification.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Ebrahimi F, Rastegar S. Preservation of mango fruit with guar-based edible coatings enriched with *Spirulina platensis* and *Aloe vera* extract during storage at ambient temperature. *Sci. Hortic* 2020;265:109258. <https://doi.org/10.1016/j.scienta.2020.109258>.
- [2] Eshetu A, Ibrahim A, Forsido S, Kuyu C. Effect of beeswax and chitosan treatments on quality and shelf life of selected mango (*Mangifera indica* L.) cultivars. *Heliyon* 2019;5(1):e01116. <https://doi.org/10.1016/j.heliyon.2018.e01116>.
- [3] FAO. (2020). *Principales frutas tropicales - Compendio estadístico 2018*. Roma, Italia. <https://openknowledge.fao.org/server/api/core/bitstreams/adc73bc5-c3ff-43a9-91cd-d153de068d40/content>.
- [4] Masibo M, He Q. Major mango polyphenols and their potential significance to human health. *Compr. Rev. Food Sci. Food Saf.* 2008;7(4):309–19. <https://doi.org/10.1111/j.1541-4337.2008.00047.x>.
- [5] Ramírez J, Cabrera C, Prieto F. A computer vision system for early detection of anthracnose in sugar mango (*Mangifera indica*) based on UV-A illumination. *Inf. Process. Agric.* 2023;10(2):204–15. <https://doi.org/10.1016/j.inpa.2022.02.001>.
- [6] Galán V. Current situation and future prospects of worldwide mango production and market. *Glob. Conf. Augment. Prod. Util. Mango: Biot. Abiotic Stress. ISHS Acta Hortic.* 2011;1066(7):69–84. <https://doi.org/10.17660/ActaHortic.2015.1066.7>.
- [7] Datir S, Regan S. Advances in physiological, transcriptomic, proteomic, metabolomic, and molecular genetic approaches for enhancing mango fruit quality. *J. Agric. Food Chem* 2022;71(1):20–34. <https://doi.org/10.1021/acs.jafc.2c05958>.
- [8] Maldonado-Celis M, Yahia E, Bedoya R, Landázuri P, Loango N, Aguilón J, Restrepo B, Guerrero C. Chemical composition of mango (*Mangifera indica* L.) fruit: nutritional and phytochemical compounds. *Front Plant Sci* 2019;10(1073):1–21. <https://doi.org/10.3389/fpls.2019.01073>.
- [9] Shaikh T, Rasool T, Lone F. Towards leveraging the role of machine learning and artificial intelligence in precision agriculture and smart farming. *Comput. Electron. Agric.* 2022;198:107–19. <https://doi.org/10.1016/j.compag.2022.107119>.
- [10] National Mango Board. (2020). *Protocolo para el manejo y la maduración del mango*. <https://www.mango.org>.
- [11] Wongkaew M, Kittiwachana S, Phuangsaijai N, Tinpovong B, Tiayon C, Pusadee T, Sringarm K, Bhat F, Sommano S, Cheewangkoon R. Fruit characteristics, peel nutritional compositions, and their relationships with mango peel pectin quality. *Plants* 2021;10(6):2–20. <https://doi.org/10.3390/plants10061148>.
- [12] Aragón D. Estudio de tendencias de mercado – mango. Lima-Perú: Instituto Nacional de Innovación Agraria; 2022. <https://hdl.handle.net/20.500.12955/2025>.
- [13] Gupta A, Pathak U, Tongbram T, Medhi M, Terdwongworakul A, Magwaza L, Mditshwa A, Chen T, Mishra P. Emerging approaches to determine maturity of citrus fruit. *Crit Rev Food Sci Nutr* 2022;62(19):5245–66. <https://doi.org/10.1080/10408398.2021.1883547>.
- [14] Salazar-Campos O, Salazar-Campos J, Menacho D, Morales D, Aredo V. Improvement of the classification of green asparagus using a computer vision system. *Braz. J. Food Technol.* 2019;22:1–5. <https://doi.org/10.1590/1981-6723.14018>.
- [15] Worasawate D, Sakunasinha P, Chiangga S. Automatic classification of the ripeness stage of mango fruit using a machine learning approach. *AgriEngineering* 2022;4(1):32–47. <https://doi.org/10.3390/agriengineering4010003>.
- [16] Selvakumar A, Balasundaram A. Automated mango leaf infection classification using weighted and deep features with optimized recurrent neural network concept. *Imaging Sci. J.* 2023;72(3):1–19. <https://doi.org/10.1080/13682199.2023.2204036>.
- [17] Lin G, Tang Y, Zou X, Cheng J, Xiong J. Fruit detection in natural environment using partial shape matching and probabilistic Hough transform. *Precis. Agric.* 2020;21:160–77. <https://doi.org/10.1007/s11119-019-09662-w>.
- [18] Wan P, Toudeshki A, Tan H, Ehsani R. A methodology for fresh tomato maturity detection using computer vision. *Comput. Electron. Agric.* 2018;146:43–50. <https://doi.org/10.1016/j.compag.2018.01.01>.
- [19] Sengupta S, Lee W. Identification and determination of the number of immature green citrus fruit in a canopy under different ambient light conditions. *Biosyst. Eng.* 2014;117:51–61. <https://doi.org/10.1016/j.biosystemseng.2013.07.007>.
- [20] Dhanush G, Khatri N, Kumar S, Shukla K. A comprehensive review of machine vision systems and artificial intelligence algorithms for the detection and harvesting of agricultural produce. *Sci. Afr.* 2023:e0179. <https://doi.org/10.1016/j.sciaf.2023.e01798>.

- [21] Zhang Y, Wu L. Classification of fruits using computer vision and a multiclass support vector machine. *Sensors* 2012;12(9):12489–505. <https://doi.org/10.3390/s120912489>.
- [22] Ratprakhon K, Neubauer W, Riehn K, Fritsche J, Rohn S. Developing an automatic color determination procedure for the quality assessment of mangos (*Mangifera indica* L.) using a CCD camera and color standards. *Foods* 2020;9(11):1709. <https://doi.org/10.3390/foods9111709>.
- [23] Li W, He D, Du X. Advances in agricultural product classification using deep learning techniques: a review. *Comput. Electron. Agric.* 2021;187:106267. <https://doi.org/10.1016/j.compag.2021.106267>.
- [24] LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature* 2015;521(7553):436–44. <https://doi.org/10.1038/nature14539>.
- [25] Zhang Y, Kang B, Hooi B, Yan S, Feng J. Deep long-tailed learning: a survey. *IEEE Trans Pattern Anal Mach Intell* 2023;45(9):10795–816. <https://doi.org/10.1109/TPAMI.2023.3268118>.
- [26] Liu Y, Pu H, Sun D. Efficient extraction of deep image features using convolutional neural network (CNN) for applications in detecting and analysing complex food matrices. *Trends Food Sci. Technol.* 2021;113:193–204. <https://doi.org/10.1016/j.tifs.2021.04.042>.
- [27] Krichen M. Convolutional neural networks: a survey. *Computers* 2023;12(8):151. <https://doi.org/10.3390/computers12080151>.
- [28] Tang Y, Qiu J, Zhang Y, Wu D, Cao Y, Zhao K, Zhu L. Optimization strategies of fruit detection to overcome the challenge of unstructured background in field orchard environment: a review. *Precis. Agric.* 2023;24(4):1183–219. <https://doi.org/10.1007/s11119-023-10009-9>.
- [29] INACAL. Norma técnica peruana 011.025:2023 - MANGO. mango fresco. determinación de madurez fisiológica (Segunda edición). Lima: Instituto Nacional de la Calidad; 2020. <https://servicios.inacal.gob.pe/cidaleta/biblioteca-detalle.aspx?id=37294>.
- [30] Bures S, Urrestarazu M, Kotiranta S. Iluminación artificial en agricultura. *Bibl. Hortic.* 2018 <https://www.poscosecha.com/biblioteca/publicaciones/iluminacion-artificial-en-agricultura>.
- [31] Payne A, Walsh K, Subedi P, Jarvis D. Estimation of mango crop yield using image analysis–segmentation method. *Comput. Electron. Agric.* 2013;91:57–64. <https://doi.org/10.1016/j.compag.2012.11.009>.
- [32] Microsoft. LifeCam HD-3000. Microsoft Accessories; 2023. <https://www.microsoft.com/en/accessories/products/webcams/lifecam-hd-3000>.
- [33] Krizhevsky A, Sutskever I, Hinton GE. ImageNet classification with deep convolutional neural networks. *Commun ACM* 2012;60(6):84–90. <https://doi.org/10.1145/3065386>.
- [34] Aldoseri A, Al-Khalifa K, Hamouda A. Re-thinking data strategy and integration for artificial intelligence: concepts, opportunities, and challenges. *Appl. Sci.* 2023;13(12):7082. <https://doi.org/10.3390/app13127082>.
- [35] de Araújo A, Azcarate S, Španik I, Khvalbota L, Goicoechea C. Pattern recognition techniques in food quality and authenticity: a guide on how to process multivariate data in food analysis. *TrAC Trends Anal. Chem.* 2023;164:117105. <https://doi.org/10.1016/j.trac.2023.117105>.
- [36] Liu W, Wang Z, Liu X, Zeng N, Liu Y, Alsaadi E. A survey of deep neural network architectures and their applications. *Neurocomputing* 2017;234:11–26. <https://doi.org/10.1016/j.neucom.2016.12.038>.
- [37] Python Software Foundation. Python (Version 3.11). Python Software Foundation; 2022. <https://www.python.org>.
- [38] Carneiro T, Da Nóbrega M, Nepomuceno T, Bian B, De Albuquerque C, Reboucas Filho P. Performance analysis of Google Colaboratory as a tool for accelerating deep learning applications. *IEEE Access* 2018;6:61677–85. <https://doi.org/10.1109/ACCESS.2018.2874767>.
- [39] Guo C, Pleiss G, Sun Y, Weinberger K. On calibration of modern neural networks. In: *International Conference on Machine Learning* 2017, PMLR. 70; 2017. p. 1321–30. <https://proceedings.mlr.press/v70/guo17a.html>.
- [40] Zhang Z, Sabuncu M. Generalised cross entropy loss for training deep neural networks with noisy labels. *Adv Neural Inf Process Syst* 2018;31. <https://proceedings.neurips.cc/paper/2018/hash/f2925f97bc13ad2852a7a551802fee0-Abstract.html>.
- [41] Xiao N, Sun L, Zhang S, Sra S. Adam-family methods for nonsmooth optimisation with convergence guarantees. *J. Mach. Learn. Res.* 2024;25(48):1–53. <https://www.jmlr.org/papers/v25/23-0576.html>.
- [42] Sokolova M, Lapalme G. A systematic analysis of performance measures for classification tasks. *Inf Process Manag* 2009;45(4):427–37. <https://doi.org/10.1016/j.ipm.2009.03.002>.
- [43] Chicco D, Jurman G. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genom.* 2020;21(1):6. <https://doi.org/10.1186/s12864-019-6413-7>.
- [44] Zhang Y, Dong Z, Chen X, Jia W, Du S, Muhammad K, Wang S. Image-based fruit category classification by 13-layer deep convolutional neural network and data augmentation. *Multimed Tools Appl* 2019;78:3613–32. <https://doi.org/10.1007/s11042-017-5243-3>.
- [45] Kumar A, Joshi R, Dutta M, Jonak M, Burget R. Fruit-CNN: an efficient deep learning-based fruit classification and quality assessment for precision agriculture. In: *2021 13th International Congress on Ultra Modern Telecommunications and Control Systems and Workshops (ICUMT)*. IEEE; 2021. p. 60–5. <https://doi.org/10.1109/ICUMT54235.2021.9631643>.
- [46] Wang S, Chen Y. Fruit category classification via an eight-layer convolutional neural network with parametric rectified linear unit and dropout technique. *Multimed Tools Appl* 2020;79:15117–33. <https://doi.org/10.1007/s11042-018-6661-6>.
- [47] Ma X, Tong J, Huang W, Lin H. Characteristic mango price forecasting using combined deep-learning optimization model. *PLOS One* 2023;18(4):e0283584. <https://doi.org/10.1371/journal.pone.0283584>.
- [48] Dong C, Loy C, Tang X. Accelerating the superresolution convolutional neural network. In: *Computer Vision–ECCV 2016, 14th European Conference, Amsterdam, The Netherlands, October 2016*. Springer International Publishing; 2016. p. 391–407. https://doi.org/10.1007/978-3-319-46475-6_25.
- [49] Wu B, Iandola F, Jin P, Keutzer K. SqueezeDet: unified, small, low power fully convolutional neural networks for real-time object detection for autonomous driving. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*; 2017. p. 129–37. <https://doi.org/10.48550/arXiv.1612.01051>.
- [50] Su J, Vargas D, Sakurai K. One pixel attack for fooling deep neural networks. *IEEE Trans. Evol. Comput.* 2019;23(5):828–41. <https://doi.org/10.1109/TEVC.2019.2890858>.
- [51] Xiao F, Wang H, Xu Y, Zhang R. Fruit detection and recognition based on deep learning for automatic harvesting: an overview and review. *Agronomy* 2023;13(6):1625. <https://doi.org/10.3390/agronomy13061625>.
- [52] Sakib, S., Ashrafi, Z., & Siddique, M. (2019). Implementation of fruits recognition classifier using convolutional neural network algorithm for observation of accuracies for various hidden layers. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.1904.00783>.
- [53] Bhargava A, Bansal A. Fruits and vegetables quality evaluation using computer vision: a review. *J. King Saud Univ.-Comput. Inf. Sci.* 2021;33(3):243–57. <https://doi.org/10.1016/j.jksuci.2018.06.002>.
- [54] Casado-García A, Heras J, Milella A, Marani R. Semisupervised deep learning and low-cost cameras for the semantic segmentation of natural images in viticulture. *Precis. Agric.* 2022;23(6):2001–26. <https://doi.org/10.1007/s11119-022-09929-9>.
- [55] Sa I, Ge Z, Dayoub F, Upcroft B, Perez T, McCool C. Deepfruits: a fruit detection system using deep neural networks. *Sensors* 2016;16(8):1222. <https://doi.org/10.3390/s16081222>.
- [56] Javadi M, Haleem A, Khan IH, Suman R. Understanding the potential applications of artificial intelligence in agriculture sector. *Adv. Agrochem* 2023;2(1):15–30. <https://doi.org/10.1016/j.aac.2022.10.001>.
- [57] Rizwan M, Hakim A. Classification and grading of harvested mangoes using convolutional neural network. *Int. J. Fruit Sci.* 2022;22(1):95–109. <https://doi.org/10.1080/15538362.2021.2023069>.
- [58] Gururaj N, Vinod V. Deep grading of mangoes using convolutional neural network and computer vision. *Multimed Tools Appl* 2023;82(25):39525–50. <https://doi.org/10.1007/s11042-021-11616-2>.
- [59] Xu W, Zhao L, Li J, Shang S, Ding X, Wang T. Detection and classification of tea buds based on deep learning. *Comput. Electron. Agric.* 2022;192:10168–1699. <https://doi.org/10.1016/j.compag.2021.10654>.
- [60] Zoph B, Vasudevan V, Shlens J, Le Q. Learning transferable architectures for scalable image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*; 2018. p. 8697–710. <https://doi.org/10.48550/arXiv.1707.07012>.

- [61] Nithya R, Santhi B, Manikandan R, Rahimi M, Gandomi H. Computer vision system for mango fruit defect detection using deep convolutional neural network. *Foods* 2022;11(21):3483. <https://doi.org/10.3390/foods11213483>.
- [62] Tai D, Lin C, Trieu M, Thinh T. Development of a mango-grading and-sorting system based on external features, using machine learning algorithms. *Agronomy* 2024;14(4):831. <https://doi.org/10.3390/agronomy14040831>.
- [63] Chicco D, Tötsch N, Jurman G. The Matthews correlation coefficient (MCC) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation. *BioData Min* 2021;14:1–22. <https://doi.org/10.1186/s13040-021-00244-z>.
- [64] Maxwell A, Warner T, Guillén L. Accuracy assessment in convolutional neural network-based deep learning remote sensing studies—Part 1: literature review. *Remote Sens (Basel)* 2021;13(13):2450. <https://doi.org/10.3390/rs13132450>.
- [65] Al Riza D, Rulin C, Tun N, Yi P, Thwe A, Myint K, Kondo N. Mango (*Mangifera indica* cv. Sein Ta Lone) ripeness level prediction using color and textural features of combined reflectance-fluorescence images. *J. Agric. Food Res.* 2023;11:100477. <https://doi.org/10.1016/j.jafr.2022.100477>.
- [66] Chen B, Zou X, Zhang Y, Li J, Li K, Xing J, Tao P. LEFormer: a hybrid CNN-transformer architecture for accurate lake extraction from remote sensing imagery. In: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE; 2024. p. 5710–4. <https://doi.org/10.1109/ICASSP48485.2024.10446785>.
- [67] Admass W, Munaye Y, Bogale G. Convolutional neural networks and histogram-oriented gradients: a hybrid approach for automatic mango disease detection and classification. *Int. J. Inf. Technol.* 2024;16(2):817–29. <https://doi.org/10.1007/s41870-023-01605-z>.
- [68] Badshah A, Alkazemi BY, Din F, Zamli KZ, Haris M. Crop classification and yield prediction using robust machine learning models for agricultural sustainability. *IEEE Access* 2024.
- [69] Li Z, Liu F, Yang W, Peng S, Zhou J. A survey of convolutional neural networks: analysis, applications, and prospects. *IEEE Trans Neural Netw Learn Syst* 2021; 33(12):6999–7019. <https://doi.org/10.1109/TNNLS.2021.3084827>.
- [70] Zhou X, Hu X, Sun J. A review of fruit ripeness recognition methods based on deep learning. *Cyber-Phys. Syst.* 2025:1–35. <https://doi.org/10.1080/23335777.2025.2467639>.
- [71] Wang, D., & Felicetti, M. (2023). Stochastic configuration machines for industrial artificial intelligence. *arXiv preprint arXiv:2308.13570*. <https://doi.org/10.48550/arXiv.2308.13570>.
- [72] Yan P, Abdulkadir A, Luley P, Rosenthal M, Schatte G, Grewe B, Stadelmann T. A comprehensive survey of deep transfer learning for anomaly detection in industrial time series: methods, applications, and directions. *IEEE Access* 2024. <https://doi.org/10.48550/arXiv.2308.13570>.
- [73] Zhang A, Ren W, Liu Y, Cao X. Lightweight image super-resolution with superpixel token interaction. In: Proceedings of the IEEE/CVF international conference on computer vision; 2023. p. 12728–37. https://openaccess.thecvf.com/content/ICCV2023/papers/Zhang_Lightweight_Image_Super-Resolution_with_Superpixel_Token_Interaction_ICCV_2023_paper.pdf.
- [74] Gorriz, J., Segovia, F., Ramirez, J., Ortiz, A., & Suckling, J. (2024). Is K-fold cross validation the best model selection method for Machine learning?. *arXiv preprint arXiv:2401.16407*. <https://doi.org/10.48550/arXiv.2401.16407>.
- [75] Jara-Vélez J, Siche R, Velásquez-Barreto F, Salazar-Campos O, Lopez Y, Salazar-Campos J. Analytical optimisation of eco-friendly soap production using hyperspectral imaging and chemometric modelling of physicochemical properties. *Microchem. J.* 2025;2025:114259. <https://doi.org/10.1016/j.microc.2025.114259>.